
I. Single choice questions (20 points in total, each 2 points)

1. The DTFT of a real odd sequence is (**D**).
- (A) Real even sequence (B) Real odd sequence
(C) Imaginary even sequence (D) Imaginary odd sequence
2. The result of the 5-point circular convolution of sequence $x[n] = \{\underset{\uparrow}{1}, 2, 0, 1\}$ and sequence $y[n] = \{\underset{\uparrow}{1}, 2, 1, 2\}$ is (**A**).
- (A) $\{\underset{\uparrow}{2}, 6, 5, 5, 6\}$ (B) $\{\underset{\uparrow}{6}, 2, 6, 5, 5\}$ (C) $\{\underset{\uparrow}{5}, 6, 2, 6, 5\}$ (D) $\{\underset{\uparrow}{5}, 5, 6, 2, 6\}$
3. Among the following finite-length unit impulse responses, the linear-phase filter is (**A**).
- (A) $h[n] = 3\delta[n-1] + 2\delta[n-2] + 3\delta[n-3]$ (B) $h[n] = 3\delta[n] - 2\delta[n-1] - 3\delta[n-2]$
(C) $h[n] = 3\delta[n] + 2\delta[n-1] + 3\delta[n-3]$ (D) $h[n] = 3\delta[n] + 2\delta[n-2] - 3\delta[n-3]$
4. The number of complex additions required to compute an N-point FFT is proportional to (**B**).
- (A) N (B) $N \log_2 N$ (C) N^2 (D) N^3
5. There are four discrete-time systems as follows: ① $y[n] = \sum_{m=-3}^3 x[n+m]$, ② $y[n] = (n+1)x[n]$, ③ $y[n] = x[-5-n]$ 和 ④ $y[n] = x[n] + 4$, The correct statement is (**C**).
- (A) system ① is nonlinear and time-invariant
(B) system ② is nonlinear and time-variant
(C) system ③ is linear and time-variant
(D) system ④ is linear and time-invariant
6. Given the impulse response $h[n]$, $0 \leq n \leq N-1$ of a minimum-phase system, the characteristic of the z-transform of the sequence $g[n] = h[N-n+1]$, $0 \leq n \leq N-1$ is

(**D**).

- (A) all zeros and poles are inside the unit circle
- (B) all zeros and poles are outside the unit circle
- (C) all zeros are inside the unit circle and all poles are outside the unit circle
- (D) all zeros are outside the unit circle and all poles are inside the unit circle

7. A linear-phase FIR filter of length 8, realized in the direct form II structure. The minimum number of delay units and multipliers required are (**D**).

- (A) 8 delay units, 8 multipliers
- (B) 8 delay units, 4 multipliers
- (C) 7 delay units, 7 multipliers
- (D) 7 delay units, 4 multipliers

8. Given the system function $H(z) = \frac{1-a^{-1}z^{-1}}{1-az^{-1}}$ (a is a real number) of a linear time-invariant causal system, if the system is stable, the range of a is (**D**).

- (A) $|a| < 1$
- (B) $0 < |a| < 2$
- (C) $0 < |a|$
- (D) $0 < |a| < 1$

9. There are four main types of linear-phase FIR filters, which are

- (I) $h[n]$ is even symmetry, the length N is odd
- (II) $h[n]$ is even symmetry, the length N is even
- (III) $h[n]$ is odd symmetry, the length N is odd
- (IV) $h[n]$ is odd symmetry, the length N is even

Among them, which one that **can** be used to design band-stop filter is (**A**).

- (A) I
- (B) II
- (C) III
- (D) IV

10. Regarding the analog frequency Ω and the digital frequency ω , the incorrect statement is (**D**).

- (A) $\Omega = \tan\left(\frac{\omega}{2}\right)$ is a nonlinear mapping relationship
- (B) $\Omega T = \omega$ is a linear mapping relationship
- (C) $\Omega = \tan\left(\frac{\omega}{2}\right)$ maps the $j\Omega$ axis of the s-plane one-to-one onto the unit circle of the z-plane
- (D) $\Omega = \tan\left(\frac{\omega}{2}\right)$ maps the left half of the s-plane outside the unit circle of the z-plane.

II. Blank-fill questions

1. The impulse response of a discrete causal LTI system is $h[n] = \left(\frac{1}{3}\right)^n u[n]$.

The difference equation of this LTI system is $y[n] = x[n] + \frac{1}{3}y[n-1]$.

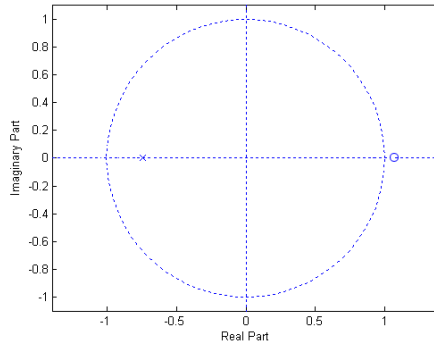
2. Given a 9-point real sequence with DFT values of $X[0] = 23, X[1] = 2.2426 - j, X[4] = -6.3749 + 4.1212j, X[6] = 6.5 + 2.5891j, X[7] = -4.1527 - 0.2645j$ at five points, determine the DFT values at the remaining 2 points:

$X[2] =$ $-4.1527 + 0.2645j$, $X[8] =$ $2.2426 + j$.

3. $X(e^{j\omega})$ is the DTFT of $\{x[n]\} = \left\{ \underset{\uparrow}{1} \text{ 噢噢噢噢 } \right\}$, Sampling $X(e^{j\omega})$ at 6 frequency points in one period to get $X_1(k)$, and taking the 6-point IDFT of $X_1(k)$ to get $y_1(n) =$ $\left\{ \underset{\uparrow}{1} \text{ 噢噢噢噢 } \right\}$. Sampling $X(e^{j\omega})$ at 4 frequency points in one period to get $X_2(k)$, and taking the 4-point IDFT of $X_2(k)$ to get $y_2(n) =$ $\left\{ \underset{\uparrow}{0} \text{ 噢噢噢 } \right\}$.

4. There is a minimum-phase filter with the same magnitude response as the filter $H(z) = (1 + 1.5z^{-1})(1 - 0.5z^{-1})$. Its transfer function is $(1.5 + z^{-1})(1 - 0.5z^{-1})$.

5. Given the pole-zero plot of a causal LTI system as shown below, the filter is a high-pass (low-pass / high-pass / band-pass / band-stop) filter, and it is maximum phase (maximum phase / minimum phase) system.



6. In commonly used analog prototype lowpass filters, the magnitude response of Type 1 Chebyshev filter and Elliptic filter has equiripple in the passband.

III. Calculations

1. A continuous-time signal

$$x(t) = 3 \sin(32\pi t) + 3 \cos(24\pi t) + 8 \sin(120\pi t) + 4 \cos(144\pi t) \text{ is}$$

sampled at a frequency of 40 Hz to obtain a discrete-time sequence,

which is then passed through an ideal lowpass filter with a cutoff at $\omega_c = 0.7\pi$.

(1) Determine the expression for the discrete-time signal obtained.

(2) Determine the fundamental period of this signal.

(3) Calculate the average power of this signal.

Answer:

(1) After sampled at a frequency of 40 Hz

$$x(n) = 3 \sin\left(\frac{32\pi n}{40}\right) + 3 \cos\left(\frac{24\pi n}{40}\right) + 8 \sin\left(\frac{120\pi n}{40}\right) + 4 \cos\left(\frac{144\pi n}{40}\right)$$

$$x(n) = 3 \sin\left(\frac{4\pi n}{5}\right) + 3 \cos\left(\frac{3\pi n}{5}\right) + 8 \sin(3\pi n) + 4 \cos\left(\frac{2\pi n}{5}\right)$$

$$= 3 \sin\left(\frac{4\pi n}{5}\right) + 3 \cos\left(\frac{3\pi n}{5}\right) + 4 \cos\left(\frac{2\pi n}{5}\right)$$

2 Points

passed through an ideal lowpass filter

$$x_1(n) = 3 \cos\left(\frac{3\pi n}{5}\right) + 4 \cos\left(\frac{2\pi n}{5}\right)$$

3 Points

(2) The period of $\cos(\frac{3\pi n}{5})$ is 10. The period of $\cos(\frac{3\pi n}{5})$ is 5, So the period of

$$x_1(n) = 3\cos(\frac{3\pi n}{5}) + 4\cos(\frac{2\pi n}{5}) \text{ is } 10. \quad 3 \text{ Points}$$

$$(3) P = \frac{1}{N} \sum_{n=0}^{N-1} |x_1(n)|^2 = \sum_{k=0}^{N-1} |a_k|^2 = 12.5 \quad 2 \text{ Points}$$

2. Let the sequence $x[n]$ of length N be padded with $(M-1)N$ zeros, resulting in an MN -point DFT $Y[k]$. Prove that the N -point DFT $X[k]$ of $x[n]$ can be obtained from $Y[k]$ by the following formula.

$$X[k] = Y[Mk], \quad 0 \leq k \leq N-1$$

Prove:

$$Y[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/MN}, k = 0, 1, \dots, MN-1 \quad 4 \text{ Points}$$

$$Y[Mk] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi Mk n/MN} = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N} = X[k], \quad k = 0, 1, \dots, N-1 \quad 4 \text{ Points}$$

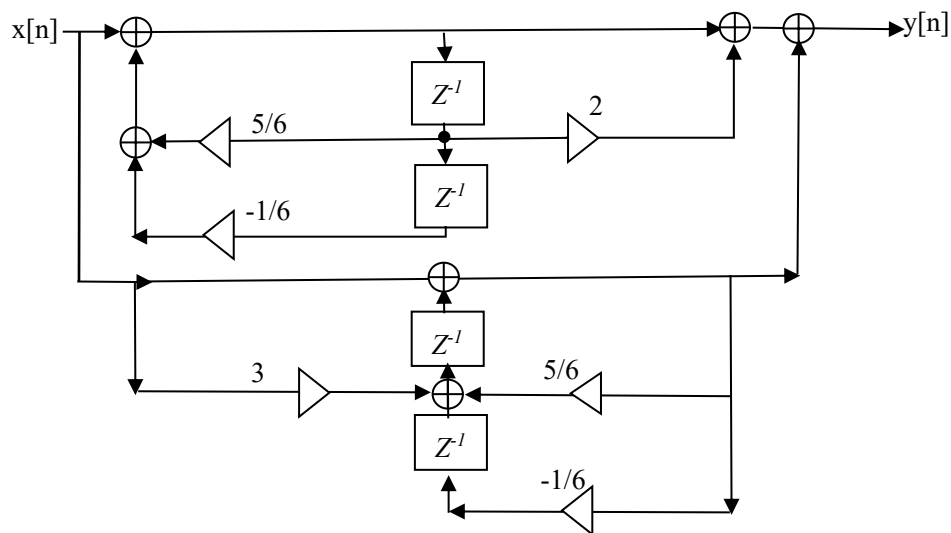
3. Given a causal system with transfer function $H(z)$ as

shown in the figure, determine the following:

(1) Write the transfer function of the system.

(2) Is the structure shown in the figure canonic? If not, develop a canonic direct-form realization.

(3) Is the system a minimum-phase system? If not, try described this system as a cascade of a minimum-phase system and an allpass system.



Answer:

$$H(z) = \frac{1+2z^{-1}}{1-\frac{5}{6}z^{-1}+\frac{1}{6}z^{-2}} + \frac{1+3z^{-1}}{1-\frac{5}{6}z^{-1}+\frac{1}{6}z^{-2}}$$

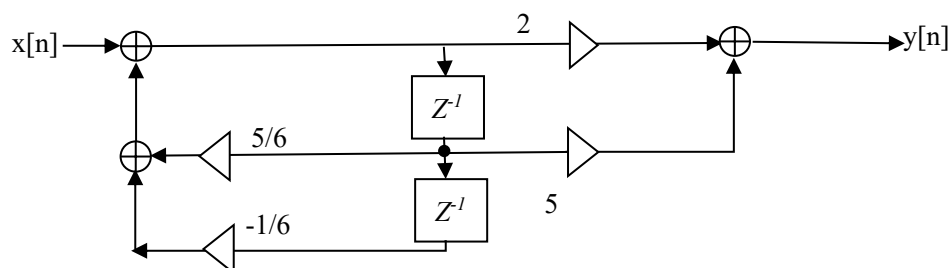
$$H(z) = \frac{2+5z^{-1}}{1-\frac{5}{6}z^{-1}+\frac{1}{6}z^{-2}}$$

3points

(2) the order is 2 ,so it's noncanonic

2points

The canonic direct-form realization is



3points

(3) zero $z = -\frac{5}{2}$, it's non-minimum-phase system

$$H(z) = \frac{2+5z^{-1}}{1-\frac{5}{6}z^{-1}+\frac{1}{6}z^{-2}} = \frac{5+2z^{-1}}{1-\frac{5}{6}z^{-1}+\frac{1}{6}z^{-2}} \frac{2+5z^{-1}}{5+2z^{-1}}$$

4points

4. (10 points) A Type 1 real-coefficient FIR filter has zeros

$$z_1 = 1 + j, \quad z_2 = 0.5$$

(1) Find the transfer function $H(z)$ of the lowest order FIR filter that meets the above conditions.

(2) Determine the causal stable inverse system $G(z)$ of the above FIR filter. Is this system a linear phase system? Why?

Answer:

$$1) \quad H(z) = (1 - (1+j)z^{-1})(1 - (1-j)z^{-1})(1 - (1/(1-j))z^{-1})(1 - (1/(1+j))z^{-1})(1 - 0.5z^{-1})(1 - 2z^{-1})$$

4points

$$2) \quad H(z) = \tilde{H}(z)A(z)$$

$$= \left(1 - \left(\frac{1}{(1-j)}\right)z^{-1}\right)^2 \left(1 - \left(\frac{1}{(1+j)}\right)z^{-1}\right)^2 (1 - 0.5z^{-1})^2$$

$$\frac{(1 - (1+j)z^{-1})(1 - (1-j)z^{-1})(1 - 2z^{-1})}{(1 - (1/(1+j))z^{-1})(1 - (1/(1-j))z^{-1})(1 - 0.5z^{-1})}$$

$$\tilde{H}(z) = (1 - (1/(1-j))z^{-1})^2 (1 - (1/(1+j))z^{-1})^2 (1 - 0.5z^{-1})^2 \quad 2points$$

$$G(z) = \frac{1}{(1 - (1/(1-j))z^{-1})^2 (1 - (1/(1+j))z^{-1})^2 (1 - 0.5z^{-1})^2} \quad 2points$$

This system is an nonlinear phase system.

2points

5. (10 points) Using the bilinear transformation method with $T = 2$, design

a causal IIR digital filter with transfer function $G(z) = \frac{-z-3}{3z+1}$.

(1) Present its prototype causal analog transfer function $H(s)$.

(2) Determine whether the analog filter is stable or not?

Answer:

(1)

$$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}, T=2$$

$$z = \frac{1+s}{1-s}$$

3points

$$H_a(s) = G(z) \Big|_{z=\frac{1+s}{1-s}} = -\frac{s-2}{s+2}$$

4points

(2) G(z) pole at -1/3, So it's stable.

3points

6. (12 points) The specifications for the highpass filter are:

$$\omega_p = 0.5\pi, \quad \omega_s = 0.3\pi, \quad \delta_p = \delta_s = 0.001.$$

Select an appropriate window function from the Appendix, and design the lowest order linear-phase FIR filter.

Answer:

(1) $20\log(0.001) = -60\text{dB}$, so choose the blackman window (4points)

(2) $\Delta\omega = \omega_p - \omega_s = 0.2\pi$, so

$$M = \frac{5.56\pi}{\Delta\omega} = \frac{5.56\pi}{0.2\pi} \approx 28 \quad (2\text{points})$$

$$w[n] = \begin{cases} 0.42 + 0.5 \cos\left(\frac{\pi n}{28}\right) + 0.08 \cos\left(\frac{2\pi n}{28}\right) & -28 \leq n \leq 28 \\ 0 & \text{otherwise} \end{cases} \quad (1\text{points})$$

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$$\omega_c = (\omega_s + \omega_p) / 2 = 0.4\pi, \text{ so}$$

$$h_{HP}[n] = \begin{cases} 0.6 & n=0 \\ -\frac{\sin(0.4\pi n)}{\pi n} & |n| > 0 \end{cases} \quad (3\text{points})$$

then

$$h_t[n] = h_{LP}[n-M]w[n-M] \quad (2\text{points})$$

Appendix:

Table1 Some fixed window functions

Rectangular:	$w[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$
hann:	$w[n] = \frac{1}{2} [1 + \cos(\frac{2\pi n}{2M+1})], -M \leq n \leq M$
hamming:	$w[n] = 0.54 + 0.46 \cos(\frac{2\pi n}{2M+1}), -M \leq n \leq M$
blackman:	$w[n] = 0.42 + 0.5 \cos(\frac{2\pi n}{2M+1}) + 0.08 \cos(\frac{4\pi n}{2M+1}), -M \leq n \leq M$

Table 2 Properties of some fixed window functions

Type of window	Main lobe width Δ_{ML}	Relative Sidelobe Level $A_{s\ell}$	Minimum Stopband Attenuation	Transition Bandwidth $\Delta\omega$
Rectangular	$4\pi/(2M+1)$	13.3dB	20.9dB	$0.92\pi/M$
Hann	$8\pi/(2M+1)$	31.5dB	43.9dB	$3.11\pi/M$
Hamming	$8\pi/(2M+1)$	42.7dB	54.5dB	$3.32\pi/M$
Blackman	$12\pi/(2M+1)$	58.1dB	75.3dB	$5.56\pi/M$